

LESSON 9-3 • SESSION 2 • PRACTICE • FORM B**Equations from Situations**

Learning target – I can write an equation from a described situation and identify its variables. · 6.EE.C.9

REMEMBER

- Say what each letter means. Then **multiply** the input by the rate for one unit.
- **Add** a starting amount or fee only if the story gives one: \$5 plus \$2 each is $c = 2n + 5$.
- If one person is always 3 years older, **add**: $a = b + 3$.

WORKED EXAMPLE

A pet sitter earns \$9 each day. Write an equation for dollars e earned in d days.

Step 1 Define: d = days, e = dollars earned.

Step 2 Rate: \$9 for each day, so $e = 9d$.

Step 3 Test $d = 5$: $e = 9 \times 5 = 45$.

Answer $e = 9d$; 5 days earns \$45

WORDS unit rate (*tasa unitaria*) · multiplicative rule (*regla multiplicativa*) · additive rule (*regla aditiva*)

PRACTICE

show your work · label units

1. Write an equation for each situation.

- \$12 per hour; cost c for h hours: $c = \underline{\hspace{2cm}}$
- \$15 per ticket; cost c for t tickets: $c = \underline{\hspace{2cm}}$
- Leo is 4 years younger than his sister. Leo's age l when she is s : $l = \underline{\hspace{2cm}}$
- \$30 fee plus \$8 an hour; cost c for h hours: $c = \underline{\hspace{2cm}}$

2. **Kayak Rental.** A lake shop rents kayaks for \$11 each hour.

- Name the independent and dependent variables and their letters. _____
- Write the equation: $c = \underline{\hspace{2cm}}$
- Find the cost for 7 hours. _____ dollars

3. **Car Detailing.** Zuri earns \$21 for every car she details. Write an equation for her earnings e for n cars. Find her earnings for 7 cars.

Hint The pay repeats for every job, so multiply.

Show your work.

Answer _____ dollars

4. **Older Cousin.** Marcus is always 5 years older than his cousin. Write an equation for Marcus's age m when his cousin is c . How old is Marcus when his cousin is 12?

Hint The age gap stays the same, so add.

Show your work.

Answer _____ years

5. **Field Trip Vans.** Each van seats 14 riders. Write an equation for the riders r in v vans. Find the riders in 8 vans.

Hint Name what the school chooses, then multiply.

Show your work.

Answer _____ riders

- 6. Two Different Rules.** Graph A passes through (1, 3), (2, 6), (3, 9). Graph B passes through (1, 3), (2, 4), (3, 5). Write an equation for each. How are the rules different?

Hint Is y a multiple of x , or x plus a number?

Show your work.

Equations _____

- 7. Summer Savings.** Nadia saves \$35 from every paycheck. She had nothing saved before.

Hint The starting amount is added once.

- a. Write an equation for her savings s after p paychecks. _____
- b. Suppose she had \$50 saved before. Write the new equation. _____
- c. Use the new equation to find the savings after 6 paychecks. _____ dollars

- 8. Bounce House.** A bounce house costs a \$60 delivery fee plus \$25 an hour. Write an equation for the cost c after h hours. Find the cost for 4 hours.

Show your work.

Answer _____ dollars

FIND THE ERROR

explain the mistake, then fix it

- 9. Leila's Work.** Stickers cost \$3 each. Leila wrote $c = n + 3$ for the cost c of n stickers. Explain her mistake and fix the equation.

TEST PRACTICE

choose the best answer

- 10.** At \$14 per item, the cost c for n items is –

A $c = n + 14$ **B** $n = 14c$ **C** $c = 14n$ **D** $c = 14$

- 11.** A \$15 fee plus \$6 an hour gives –

A $c = 6h + 15$ **B** $c = 15h + 6$ **C** $c = 21h$ **D** $c = 6h$

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

112 V	77 S	260 E	160 D	147 O	17 L	18 X	340 Z	60 Q
------------------------	-----------------------	------------------------	------------------------	------------------------	-----------------------	-----------------------	------------------------	-----------------------

#2c	#3	#4	#5	#7c	#8
-----	----	----	----	-----	----

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____